POLYDEX

HOT AND COLD WATER SYSTEM



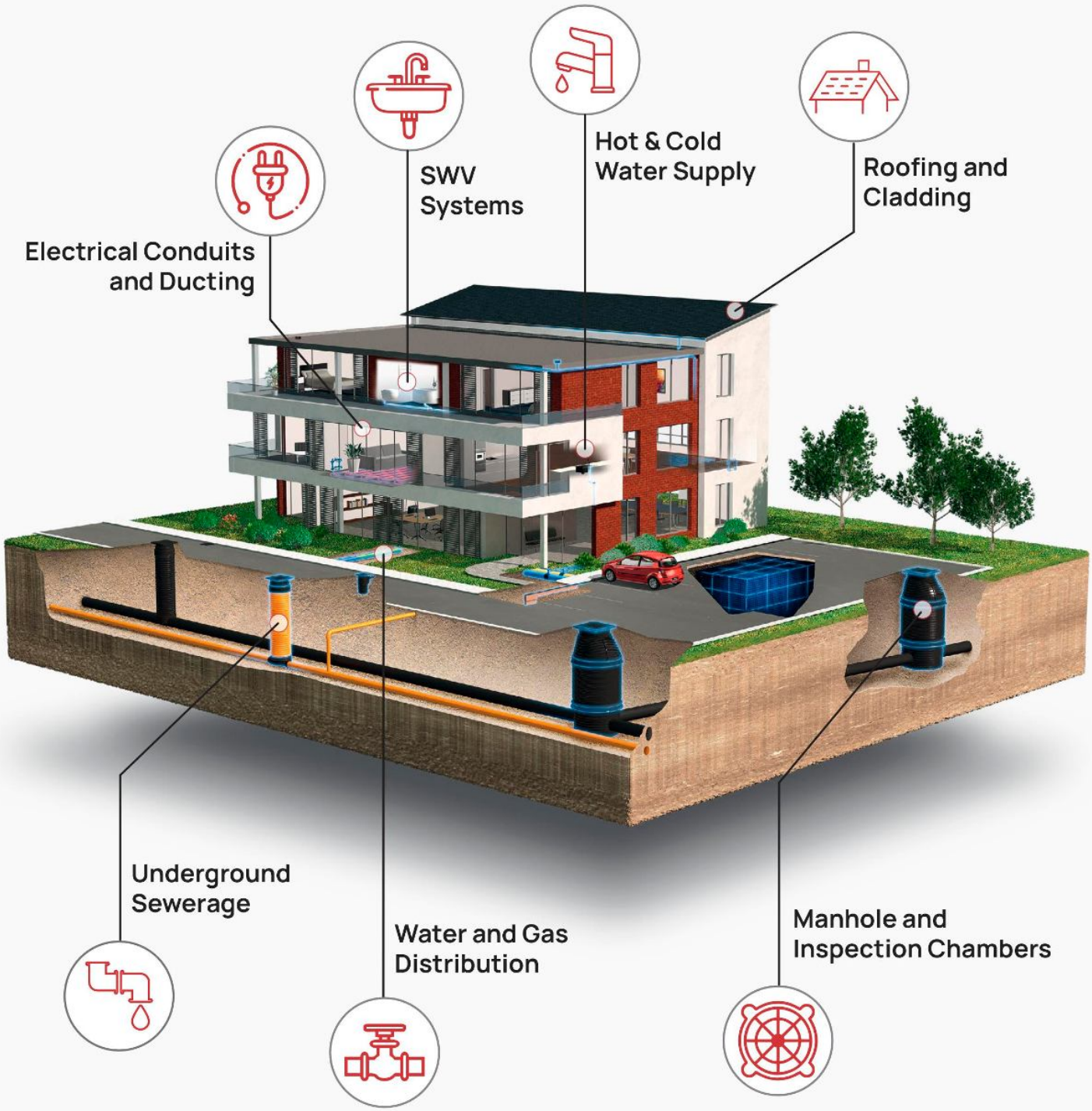


PP-R

DADEX

DADEX

Leader in Roofing and Thermoplastic Piping Systems



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ABOUT DADEX

Dadex was established as a public limited company in 1959. We started with the manufacturing of FC roofing systems at our facilities in Hyderabad, under the dynamic leadership of our Founder Chairman, Mr. Kassim Dada. Being a forward-looking organization, his son, Mr. Sikander Dada kept up to date with the market, and was instrumental in the diversification with plastic piping systems. Presently we produce various thermoplastic systems such as UPVC, PP-R, PP-RCT, PP, PE.

Headquartered in Karachi, our two manufacturing facilities based in Hyderabad and Sundar Industrial Estate (near Lahore) are equipped with state of the art machinery. Our joint efforts have led Dadex to the ISO 9001, ISO 14001, ISO 45001 and PSQCA certifications.

We aim to set standards in every field we enter. Customer focus, quality and innovation are reflected in every aspect of our business. Offering customized solutions to a cross section of markets for over 65 years, Dadex today is a hallmark of trust and reliability.

We believe that our strength lies in delighting our customers. That is why providing quality products backed by round the clock technical support services are the Dadex promise to all our valued customers.

EXTENSIVE PRODUCT RANGE

Water Supply - Infra-Structure

- Aquadex (UPVC Pressure Pipe System)
- T-Flex (PE 100 Pipes & Fittings)

Water Supply - Housing

- Polydex Premium (PP-RCT Pipe System)
- Polydex (PP-R Pipe System)
- Thermoline (PP-R Pipe System)
- Aquadex (UPVC Pressure Pipe System)

Soil, Waste and Vent (SWV)

- Nikasi - Solvent (UPVC Pipe System)
- Nikasi - Rubber Ring (UPVC Pipe System)
- Polyduct (Low Noise System)

Underground Sewerage and Drainage

- Flow Line (UPVC Underground Sewerage System)
- Inspection Chambers and Manholes

Cable Ducting

- PE Cable Duct Pipes
- Electroduct (UPVC Conduits as per IEC - EN Standard)
- Conduits (UPVC Conduits as per Dadex Standard)

Gas Supply

- T-Flex for gas (PE 80 and PE 100)

Utilisation of Dadex Polydex System

Polydex is the first complete PP-R piping system in Pakistan that conforms to international quality and standards. The Polydex piping system can be used for distribution systems in housing, administration and community buildings as well as for industrial and agricultural installations.

The system is designed for transport of cold and hot water as well as floor heating. In case of observance of the rules specified hereinafter the system is also suitable for central heating systems.

In order to take further advantage of its chemical resistance and other system properties and use the system for transport of other liquids or gaseous or solid media; such usage, however, must not be done without a prior individual assessment. Chemical disinfection of hot water must not be performed without a prior assessment by the manufacturer. Permanent disinfection of hot water with chlorine dioxide shortens the service life of the system and therefore is not recommended.



Material

Polydex pipe system is manufactured from Polypropylene Random (PPR).

The physical and chemical properties of PPR make it a superior and safe piping system.

Polydex pipes and fittings are supplied in green colour.

Threaded Fittings are available in Lead Free Brass with Silver Nickle Coating

Standards & Specifications

Polydex pipe system conforms to the following international standards:

Pipes	DIN 8077-8078 PN 20
Fittings	DIN 16962

Fields of Application

Hot and cold water installations for:

- Residences and apartments
- Hospitals
- Hotels and Offices
- Educational Buildings
- Commercial buildings and plazas
- Swimming pools

Various Industrial Applications

- Chilled water piping
- Transportation of aggressive fluids & chemical
- Transportation of food grade liquids



Features & Benefits

Potability: Polydex conforms to international standards that approve its use for distribution of drinking water.

Welding capability: All elements of Polydex pipes system can be easily joined by heat fusion.

Metal insert in fittings: All threaded inserts in the fittings are made of nickel plated brass and machined from solid bars. Polydex pipes and fittings can easily be fixed with metal fitting such as taps, bib cocks and stop cocks which enhance its utility for a wide range of applications.

Reduced head loss: Polydex has a smooth internal surface, hence there are no chances of encrustation. This leads to reduced head loss and smooth flow through out the service life.

Resistant to frost: Polydex pipe system can sustain sub zero temperatures. Elasticity of the material allows the pipes to expand if the liquid being conveyed freezes due to low temperature. Polydex does not crack or break due to frosting.

Resistant to abrasion and corrosion: Polydex resists corrosion and abrasion even in presence of acidic and alkaline substances having pH values ranging between 1 to 14.

Environment friendly: Polydex is environment friendly and can be recycled.

Low electrical conductivity: Polydex is a poor conductor of electricity and is safe against stray currents.

Low heat conductivity: The material used for Polydex is a poor conductor of heat. As a result, condensation and heat dispersion of the transported fluid are reduced. Due to this, it instantly allows hot and cold water to flow through pipes.

Fitness for use in seismic areas: Polydex is manufactured from a relatively flexible material and can sustain earth movements to an extent therefore it can be used in seismic areas.

Available Range

Polydex pipes are available in the following outside diameters and pressure classes:

Nominal OD	Min. Wall Thickness (mm)	
mm	PN 16	PN 20
20	2.8	3.4
25	3.5	4.2
32	4.4	5.4
40	5.5	6.7
50	6.9	8.3
60	8.6	10.5
75	10.3	12.5
90	12.3	15.0
110	15.1	18.3
125	17.1	20.8
160	21.9	26.6
180	24.6	29.0
200	27.4	33.2
250	34.2	-

Complete range of fittings is available. Special fittings could be made such as flange connections and short by-pass bends.

Classes of Application

The operating conditions of water distribution and heating are specified for four different classes of application (ISO 10508).

The pipes are produced in various combinations of operating pressure and temperatures in separate pressure lines (of various wall thicknesses):

PN 16 (S 3,2) for hot water and floor heating

PN 20 (S 2,5) for hot water and central heating

Each class of application refers to a typical field of use and for a period of 50 years. Each class of application must be associated with the pressure it is designed for (operating system pressure). This information is assigned to each pipe in the following format: class of application/ pressure, for example 1/10 bar. It means the pipe is categorized as class 1 and has the maximum operating pressure of 10 bar. The classes of application according to ISO 10508:

class 1 (supply of hot water of 60°C, service life 50 years)

class 2 (supply of hot water of 70°C, service life 50 years)

class 3 (floor heating, low temperature heaters, service life 50 years, assuming (in total for the entire lifetime) 2.5 years at the operating temperature of 20°C, 20 years at the operating temperature of 40°C, 25 years at the operating temperature of 60°C, 2.5 years at the operating temperature of 70°C

class 4 (high temperature heaters, service life 50 years, out of which (in total for the entire length of service life) 14 years at the operating temperature of 20°C, 25 years at the operating temperature of 60°C, 10 years at the operating temperature of 80°C, 1 year at the operating temperature 90°C). Maximum operating pressure (4, 6, 8, 10 bar) corresponding to the application class is calculated and assigned for each material and pipe series S.

Spacing distance between pipe supports

Maximum distances of supports of pipeline Polydex **PPR** (PN 16) pipe (horizontal pipeline)

ø pipe [mm]	spacing distances [in cm] at temperature of					
	20°	**30°**	**40°**	**50°**	**60°**	**80°**
16	80	75	75	70	70	60
20	90	80	80	80	70	65
25	95	95	95	90	80	75
32	110	105	105	100	95	80
40	120	120	115	105	100	95
50	135	130	125	120	115	100
63	155	150	145	135	130	115
75	170	165	160	150	145	125
90	180	180	170	165	160	135
110	200	195	190	180	175	155
125	220	215	200	195	190	165

Maximum distances of supports of pipeline Polydex **PPR** (PN 20) pipe (horizontal pipeline)

ø pipe [mm]	spacing distances [in cm] at temperature of					
	20°	**30°**	**40°**	**50°**	**60°**	**80°**
16	90	85	85	80	80	65
20	95	90	85	85	80	70
25	100	100	100	95	90	85
32	120	115	115	110	100	90
40	130	130	125	120	115	100
50	150	150	140	130	125	110
63	170	160	155	150	145	125
75	185	180	175	160	155	140
90	200	200	185	180	175	150
110	220	215	210	195	190	165
125	235	230	225	210	200	170

Maximum spacing distances between supports for vertically oriented pipelines should be multiplied by 1.3.

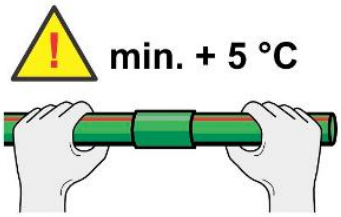
Assembly Instructions

General

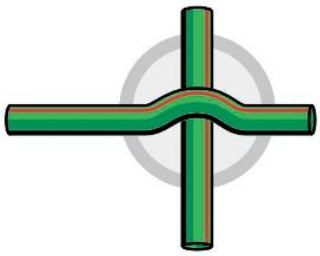
Only components not damaged or contaminated, either during storage or transport, may be used for installation works.



Components of plastic piping systems must be protected against damage during transport and installation.

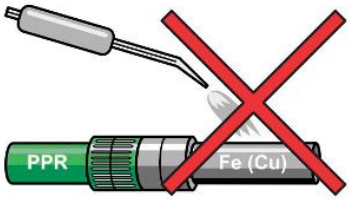


At lower temperatures it is difficult to provide proper working conditions for high-quality pipe coupling.



Pipeline cross-overs should be made with components designed explicitly for this purpose.

plastic components. Threads are sealed with a special PTFE tape or sealing compounds.



Brazing or soldering of metal fittings should not take place close to a joint between metal and plastic systems due to the potential hazard of heat transfer to the fitting.

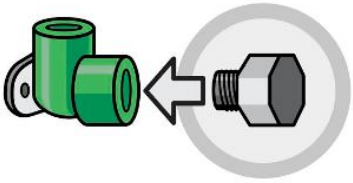


Components must not be exposed to naked flames.

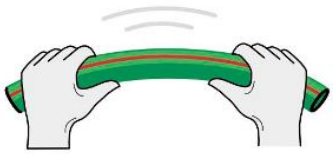


Plastic parts joining is done by polyfusion welding or by using electro-fusion sockets and butt welding techniques.

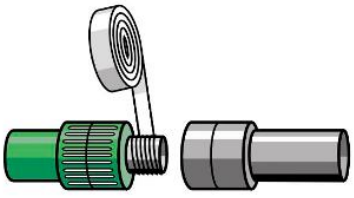
A high-quality homogeneous joint is the result. The welding process must follow proper procedures and must be performed with appropriate tools.



It is recommended to use plastic plugs for blanking elbows or wall mounting groups (plastic plugs are designated only for temporary use). Long term blanking requires using plugs with metal thread.



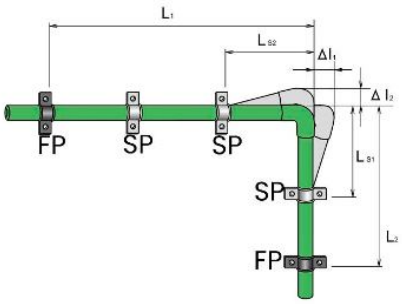
Pipe bending should be done at +15°C. For pipes of diameter range 16–32mm a minimum bending radius equals to eight diameters (D).



Threaded fittings must be used for screw-type joints. Threads should never be cut directly into

Linear expansion and Contraction

Temperature differences during installation and under service conditions (i.e. when a medium flows through the system at a different temperature than that during the course of installation) result in linear changes – expansion or contraction (Δ)



$\Delta l = a \cdot L \cdot \Delta t$ [mm]

Δl linear change [mm]

a thermal expansion coeffcient [mm/m°C] for all plastic pipes design purposes a = 0,12, for multilayer pipes respectively a = 0,05

L effective length (distance between two neighbouring fixed points in line) [m]

Δt installation and service temperature difference [°C]

$L_s = k \cdot \sqrt{(D \cdot \Delta l)}$ [mm]

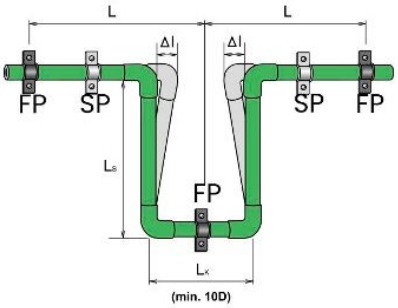
L_s compensating length [mm]

k material constant, for PPR k = 20

D external diameter of the pipe [mm]

Δl linear change [mm]

Expansion U-bend

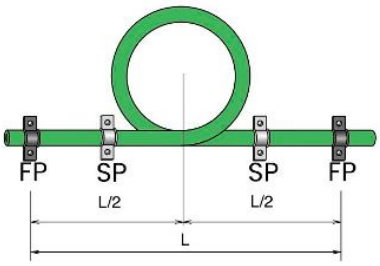


FP – fixed point
SP – sliding point
L – design pipe length
Δl – compensating length
L_s – linear change
L_k – bend width

$L_k = 2 \cdot \Delta l + 150$ [mm] and also $L_k \geq 10.D$

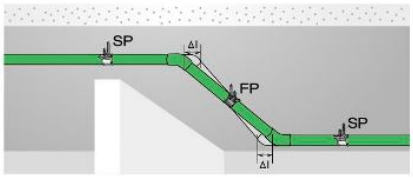
If pipeline linear changes are not compensated in a suitable way, i.e. if the pipes cannot contract and expand, additional pressure and tensile stress accumulating in the pipes shortens their service life.
In polypropylene applications linear compensation is often achieved through the pipe material's flexibility. Pipe bends can also be used for these purposes. One suitable compensation technique is where the pipeline is deflected perpendicularly to the original route and a free compensating length (marked as Ls) is left at the normal line. The value of Ls compensating length depends on calculated route extension (shortening), pipe material and diameter.
The values of Δl linear change and L_s compensating length can also be taken from the graphs shown on pages 20-22.

Compensation pipe

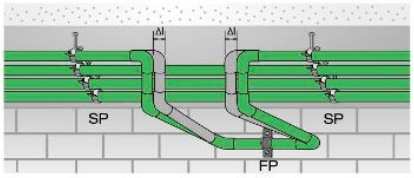


Compensation pipe table

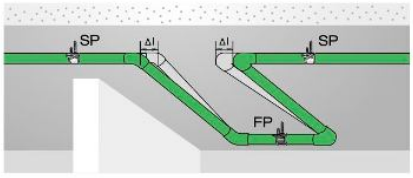
pipe dia (mm)	distances of all fixed L-points (m)	
	multilayer pipes	all plastic pipes
16	24	8
20	27	9
25	30	10
32	36	12
40	42	14



Compensation through modified pipeline routing



Compensation through modified height of the pipeline.



Compensating U-bend

Assembly Instructions

Calculation examples for Polydex PPR piping

1) Data input:

parameter	symbol	value	unit
linear change	**Δl**	**?**	**mm**
thermal expansion coefficient	*a*	0,12	mm/m °C
pipe length	L	**10**	m
operating temperature inside the pipe	t_p	60	°C
temperature at installation	t_m	20	°C
difference between operating and installation temperature levels ($\Delta t = t_p - t_m$)	Δt	**40**	°C

Solution: $\Delta l = a . L . \Delta t$ [mm]

$\Delta l = 0{,}12 . 10 . 40 =$ **48mm**

2) Data input:

parameter	symbol	value	unit
compensating lenght	**L_s**	**?**	**mm**
PPR material constant	k	20	–
external pipe diameter	D	**40**	mm
linear change as calculated above	Δl	**48**	mm

Solution: $L_s = k . \sqrt{(D . \Delta l)}$ [mm]

$L_s = 20 . \sqrt{(40 . 48)} =$ **876mm**

3) Data input:

parameter	symbol	value	unit
Compensating U-bend width?			
external pipe diameter	D	**40**	mm
linear change as calculated above	Δl	**48**	mm

Solution: $L_k = 2 . \Delta l + 150$ [mm]

$L_k = 2 . 48 + 150 = 246$ mm

$L_k \geq 10 . D$

246 mm < 10 . 40 => $L_k =$ **400 mm**

Pipe pre-stressing can be also used for the purposes of linear expansion compensation; the compensating length may be made shorter this way. The pre-stressing orientation is opposite to the expected linear change and its value equals to about one half.

4) Data input:

parameter	symbol	value	unit
compensating lenght at pre-stressing	**L_{sp}**	**?**	**mm**
PPR material constant	k	20	–
external pipe diameter	D	**40**	mm
linear change as calculated above	Δl	**48**	mm

Solution: $L_{sp} = k . \sqrt{(D . \Delta ½)}$ [mm]

$L_{sp} = 20 . \sqrt{(40 . 24)} =$ **620mm**

Compensating length Ls (Lsp) is understood as a length of pipe without any sustaining or suspending elements (on this length) obstructing expansion. The compensating length Ls (Lsp) should not exceed max. spacing distance between supporting elements corresponding to the pipe diameter and media temperature as shown in page 22.

Material Transport and Storage

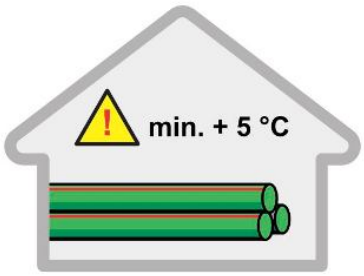
Protection

The system components must be protected against weather, UV radiation and contamination.

Stores of plastic components must be separated from such areas where solvents, adhesives, paints or similar products are stored.

It is recommended to store the components at minimum temperature of 5°C.

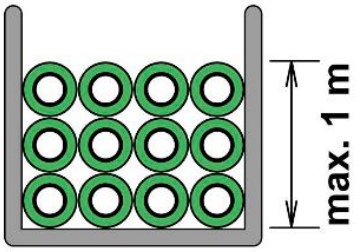
If the temperature is lower than 5°C, it is necessary to be careful at pipeline manipulation.



Plastic pipes in stock must be supported along their whole lengths or protected in another suitable way against deflection. Plastic pipe fittings are usually stored in sacks on palettes or freely loaded in boxes, containers, baskets, etc.

Maximum storage height of one metre must be respected if plastic pipes are stored in plastic sleeves and/or pipe fittings in plastic sacks. Each of the different types of pipes and fittings are stored separately. When dispatching from storage, the oldest stock should be dispatched first.

During their transport it is not allowed to drag pipes over the ground or lorry deck. The components/pipes must not be transported by throwing or letting them fall off the lorry to the ground. If they are transported to/on the site then they must be protected against mechanical damage and stored at the spot on a suitable underlay where protected against dirt, solvents, direct heat (contact with a radiator, etc.). The components are supplied in protective covers (pipes in polyethylene bags, pipe fittings also in sacks or cardboard boxes) and it is desirable to let them in these as long as possible before the start of installation works (to protect them against dirt and other contamination).



Safety welding principles

1. Welder is an electrical equipment. The condition of safe use consists of the electrical connection in manner prescribed by standard (three-wire system, in socket with earthing contact with the corresponding current load).
2. The electrical installation must be equipped with properly chosen overload protection.
3. The power cord must be protected against mechanical damage and burning of insulation cover.
4. Welder must be protected against humidity and direct contact with water.
5. Welder shall not be used in flammable and explosive environment.
6. Welder must be handled carefully, and protected against hitting, throwing, inappropriate pulling the cord.
7. In case the welder does not work properly, you must immediately unplug it from source of electric power. Damaged welder must be sent back to the manufacturer or seller together with an accurate description of the defect.
8. Operation with the welder requires special caution because of the high temperature of its body and nozzles.
9. Never leave the welder unattended while energized.
10. The welding process should al ways be performed in a well-aired room (due to the vapors formed during heating and melting of plastic elements while welding).

Polyfusion Welding Working Procedures

Tools

1. Electric polyfusion welding machine fitted with welding adapters of suitable dimensions, incl. a movable feeder cable.
2. Contact thermometer.
3. Special shears or cutter (i.e. jaw with a cutting wheel); metal hand saw if nothing else is available.
4. Sharp penknife with short blade.
5. Textile rag (non-synthetic material only).
6. Degreaser (Tangit, Spirit).
7. Folding ruler + marker.
8. Scraper and assembly jig for welding of diameters over 50 mm.

Tool check

First, attach the welding adapter to the welding machine (usually done with screws, depending on the type of the welding machine). With the welder controller set the temperature at 250 – 270°C and connect the power supply.

Time necessary to heat up the welding machine will depend on ambient conditions. Once the welding machine is warm, clean the welding adapters of impurities left from the previous use with a rag made from non-synthetic textile; failure to do so may damage the teflon surfaces. You may start working with the welding apparatus when it is sufficiently hot, which you can check using the LED diode or a contact thermometer. Use the contact thermometer if you need to fine-tune the temperature up to 250 – 270°C.

Check the proper functionality of the special shears or cutting wheel by making two cuts on a pipe assigned for testing purposes. The trial cutting should not deform or depress the outer diameter of the pipe; if it does, the tools need to be sharpened further.

Materials check

All the material should be inspected thoroughly prior to welding. The components must not contain any wall thinning defects and closing elements should be checked if they function properly and correspond to their matching piece. Welding sockets and the end of the pipe to be inserted into the fitting should be cleaned and degreased. Test the fittings by sliding them on the heating plug (which of course must remain cold for now) and check if they are not too loose. Wobbly fittings must be discarded!

Welding process

1. Measure and cut off a piece of pipe of a desirable length. When using a metal hand saw, make sure to clean the burr and dross from the pipe's edge.



2. It is also recommended to bevel the outer edges of the pipe that we mean to heat up. Use of a special cutter or a sharp knife at approximate angle of 30–45 degrees. This procedure is strongly recommended especially for pipe diameters over 40 mm. This modification will prevent the material from bunching up when the pipe end is inserted into a fitting.

3. The cleaned pipe then can be handled in the same way as the Polydex PPR pipes.



4. If you are welding larger diameters (40 mm and more) it is recommended that you check their ovality first. You should also scrape off the oxidized surface layer (about 0,1 mm in depth) along the whole length of the pipe to be inserted. The oxidized material can have negative effect on the weld quality.

Polyfusion Welding Working Procedures

⑤ The insertion depth of the pipe (to be inserted into a pipe fitting) should be marked with a marker or a sharpie.
Keep in mind that the pipe must not be pushed into the fitting socket as far as it can possibly go. About a 1mm gap should be preserved for material accumulation so as not to reduce the fitting's intern.



⑥ Furthermore, it is recommended that you also mark the weld position on both the pipe and the fitting, in order to prevent the pipe from rotating after insertion. You can also use installation marks on fittings for the same purpose.

⑦ The surfaces to be welded should be cleaned and degreased after the marking is complete. If this step is left out, then the melted layers may bond inadequately! Now you can begin the heating-up process.



⑧ Due to its thicker walls, the fitting needs more time to heat up properly and should be put on the welding adapter first. Check if it is not too loose once again; a fitting that wobbles or does not fit the whole surface of the adapter should be discarded because uneven contact (and subsequent uneven heat up of the material) may result in poor-quality welds. Next slide the pipe onto the adapter and repeat the same procedure as with the fitting.



⑨ Heat up both components for the time shown in Table, page 32. The heating time is measured starting with the moment when both the pipe and the fitting are set on the adapter with their whole marked lengths. If you experience difficulties sliding the components in place, try to rotate them slightly (max. 10°) until they are fully settled on the adapter. You may not rotate the components once the heating period begins; it could cause material to bunch up in some sections.

⑩ After the heating period is finished, remove both the fitting and the pipe and couple them together by inserting the pipe into the fitting socket for the whole marked length with gentle and uniform pressure. Again, do not rotate the components. After this is done, check the axis alignment of both components. Table on page 32 shows maximum permissible intervals between the component removal from the adapter and their coupling. Exceeding these intervals will cause the melted layers to cool excessively, thus creating a poor quality joint due to insufficient fusion of the material.

Recommendation for welding operations involving large diameters:

Pipes up to 40 mm in diameter may be welded while holding them in your hand; for the larger diameters (50mm and more) machine welding or at least a welding jig is recommended in order to ensure sufficient pressure and correct alignment of the pipes.

Preparation of pipes

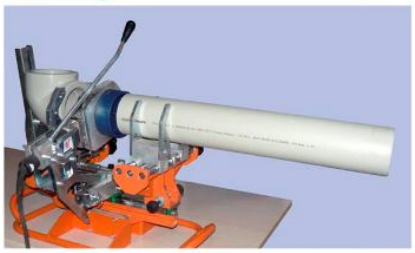


bevelling

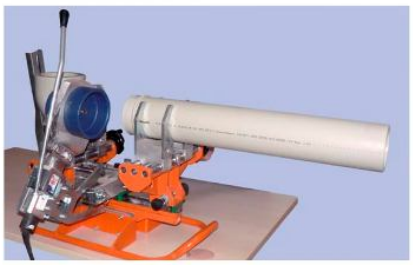


scraping

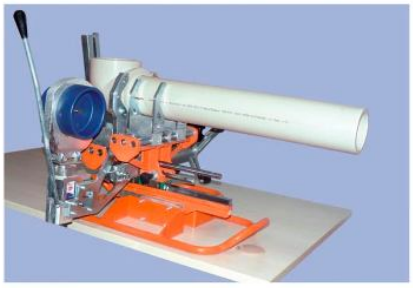
Welding



fixing and aligning the components, then heating them up



adjustment after the heating period



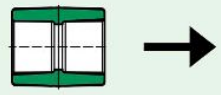
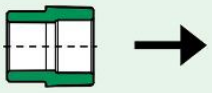
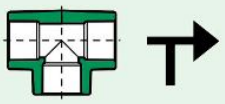
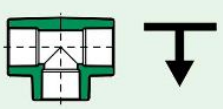
finished weld after the cool-down

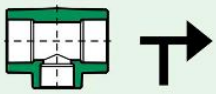
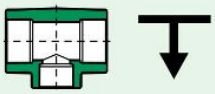
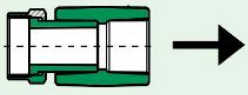
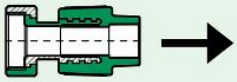
Polyfusion Welding Working Procedures

The minimum ambient temperature for welding is + 5°C

D [mm]	insertion depth L (mm)	heat-up time [s]	time for adjustment [s]	cooling time	
				fixation (s)	total (min)
16	13	5	4	6	2
20	14	5	4	6	2
25	15	7	4	10	2
32	17	8	6	10	4
40	18	12	6	20	4
50	20	18	6	20	4
63	26	24	8	30	6
75	29	30	8	30	6
90	32	40	8	40	6
110	35	50	10	50	8
125	41	60	10	60	8

Coefficient of Loss ζ

	ζ
 Socket	0,2
 Reduction by 2 dimension	0,55
 Elbow 90°	1,5
 Tee – passage in case of separation of flow	1,1
 Tee – separation of flow	1,5

	ζ
 Tee reduced – pasasage in case of flow	1,1
 Tee reduced separation of flow	4,3
 Reducing sleeve with metal thread	0,4
 Metal reducer with cap nut	8,3

Welding With Electrofusion Sockets - Working Procedures

Tools

1. Electrofusion welding machine for electric welding of plastic pipes (electrofusion socket welder)
2. Special shears or cutter.
3. Textile rag (non-synthetic material only).
4. Degreaser, sharp knife with short blade.
5. Folding ruler + marker
6. Welding jig to secure the pipe and fitting and ensure their correct alignment.
7. Scraper and assembly jig for welding of diameters over 50 mm.



Preparing the tools

Place the welding machine at your workplace and unreel the power cable. Check your cutting tools (for details see Chapter 13: Polyfusion Welding).

Welding process

Check the pipe and the fitting and prepare the welding machine.

Prepare the pipe of required length and remove the oxidized layer with a scraper or special agent. Degrease the external surface of the pipe and internal surface of the electrofusion socket (use alcohol or Tangit).

After you are done, mark the pipe insertion depth (depth of insertion inside the electrofusion socket).

Make sure the pipe is firmly fixed in the socket because due to the volume expansion of heated plastic material the pipe could be pushed out of its place during the heating process.

Connect the electrofusion welding machine to the power supply (220V) and wait until the welding machine gets into operational state. Then connect the contacts of the electrofusion socket and welding machine. Welding starts after you push the START button; the welding machine switches off automatically after the weld is completed. You can tell the electrofusion welding is proceeding well if the excess plastic material is being squeezed out of the check-points on the socket's surface. The newly coupled piping system may be filled with water not earlier than one hour after the last weld has been completed.

It is very important to let the new joint cool down before exposing it to any strain. The weld should be also protected from mechanical tension (pipe's rotation or pull).

Butt Welding Working Procedures

We recommend using a special cutter for plastic pipes.

Butt welding is one way of coupling plastic piping systems and its components together. Butt welding is a process in which two pipe ends or a pipe end and a fitting are coupled by pressing the melted contact surfaces together. Butt welding can be performed only with welding equipment specifically designed for such purpose and only by appropriately qualified staff.

Butt welding is not suitable for coupling pipes and fittings of different size (the diameter and wall thickness must be the same) or different MFI value.

The following text describes the basic welding process. Detailed welding equipment manuals including welding charts are provided by the manufacturer/supplier of the welding equipment.

In order to achieve more solid welds, we recommend using welding machines with hydraulic jaws.

The butt welding procedures described herein are based on DVS 2207.

Butt welding working instructions

Welding preparations

Workplace check

Make sure the workplace meets all the health and safety requirements and take note of dust and weather conditions. During the butt-welding process, the ambient temperature must not fall below 5 °C (use a mounting tent if needed). Similar measures must be taken in case of adverse weather conditions (rain, direct sunlight, etc.).

Welding equipment check

Check the technical condition of the welding machine (the surface and temperature of the heating plate, alignment of the fixed and movable jaws, milling cutter's functionality etc.).

Materials check

Warning: Check the mutual weldability of the materials before starting the welding process. Ensure the materials you are planning to couple have the same temperature. Butt welding can only be performed with pipes with the same wall thickness (3 mm minimum).

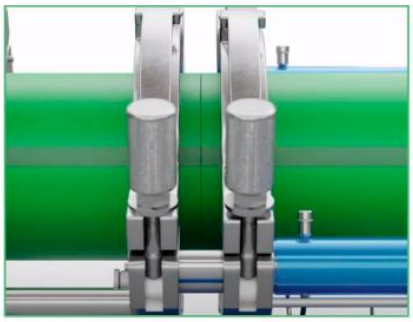
Preparing the materials

Once prepared and properly fastened, cut the pipe perpendicularly to the pipe axis with the tool designated for the purpose. Since the contact surface of the pipe must be clean, dry and free of any oils or fat, avoid using chainsaws with oil chain lubrication.

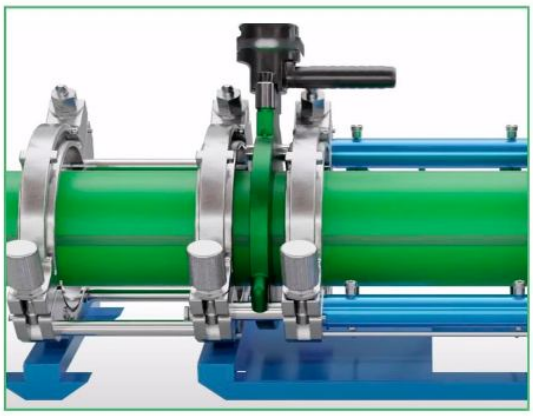
After completing the cut, always make sure to remove any burrs, sawdust or any other impurities created during the pipe cutting process. Check if the newly created contact surfaces fit well together; a careful inspection can reveal inadequate pipe ovality or tucked ends caused by the manufacturing process. Find out the passive resistance, plane the pipe surfaces as needed, test the fit once again, check for any gaps between the pipes, and finally clean the external surface of the pipes.



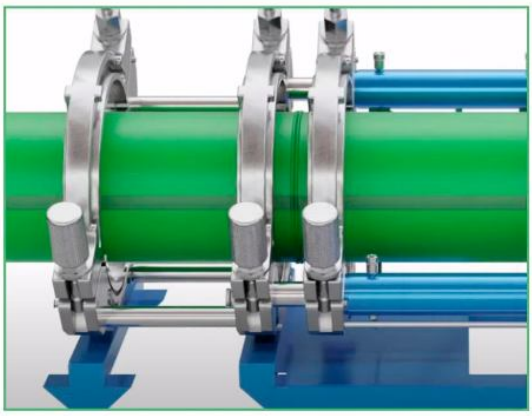
Planing the pipe ends



Properly prepared for welding



Heating with the heating plate



Cooling the weld seam

Welding phase

Truing phase

Push the contact surfaces against the heating plate until the two contact surfaces line up, which is indicated by the level of burr listed in the welding charts.

Heating phase

The contact surfaces should be heated with minimal contact pressure (see the welding charts for details). Heat up the contact surfaces until the welding zone reaches plastifcation (see the welding charts for details).

Positioning phase

Slide the contact surfaces away and remove the heating plate. Slide the pipes back as quickly as possible and push them together so their contact surfaces touch.

Coupling phase

When the contact surfaces are touching, raise the contact pressure until you reach full welding pressure.

Cooling phase

During the cooling phase keep the pipes under constant pressure (must be over-seen the welder), until the time allotted for cooling runs out. After the weld is done, perform its visual inspection in accordance with the client's requirements.

Compatibility

In order to ensure proper welding compatibility, pipes and fittings meant for butt welding should not be combined with pipes or fittings produced by a different manufacturer.

Welding parameters

Included with the welding equipment.

Guideline values for polypropylene as specified in DVS 2207 - Part 11:

Hot body temperature: 200–220 °C

Welding pressure:

- Truing and coupling: 0.10 N/mm ^{2}
- Heating: 0.01 N/mm ^{2}

Welding time: divided according to the phases - see the chart below.

1	2	3	4	5	
	Truing	**Heating**	**Positioning**	**Coupling**	
	Hot body temperature: 210 +/- 10 °C				
Nominal wall thickness	Burr level on the hot body after truing period (min. values)	(heating p. = ≤ 0.01 N / mm^2)	Positioning period (max.)	Coupling pressure rise period	Cooling under coupling pressure period (min. values) p = 0.10 N /mm^2 +/- 0.01
	(truing p = 0.10 N / mm^2)				
[mm]	**[mm]**	**[s]**	**[s]**	**[s]**	**[min]**
up to 4.5	0.5	up to 135	5	6	6
4.5 - 7	0.5	135 - 175	5 - 6	6 - 7	6 - 12
7 - 12	1.0	175 - 245	6 - 7	7 - 11	12 - 20
12 - 19	1.0	245 - 330	7 - 9	11- 17	20 - 30
19 - 26	1.5	330 - 400	9 - 11	17 - 22	30 - 40
26 - 37	2.0	400 - 485	11 - 14	22 - 32	40 - 55
27 - 50	2.0	485 - 560	14 - 17	32 - 43	55 - 70

Pressure Loss Tables

PPR S 3,2 (PN16) water temperature = 10° C

k=0,01	16 × 2,3mm		20 × 2,8mm		25 × 3,5mm		32 × 4,5mm		40 × 5,6mm		50 × 6,9mm		63 × 8,7mm		75 × 10,4mm		90 × 12,5mm		110 × 15,2mm		125 × 17,1mm	
Q l/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s
0,02	0,083	0,2	0,027	0,1	0,009	0,1																
0,04	0,282	0,4	0,093	0,2	0,032	0,2	0,010	0,1	0,003	0,1												
0,06	0,576	0,6	0,189	0,4	0,065	0,2	0,020	0,1	0,007	0,1	0,002	0,1										
0,08	0,958	0,8	0,313	0,5	0,108	0,3	0,034	0,2	0,012	0,1	0,004	0,1										
0,10	1,422	1,0	0,465	0,6	0,160	0,4	0,050	0,2	0,017	0,2	0,006	0,1	0,002	0,1								
0,12	1,967	1,2	0,641	0,7	0,221	0,5	0,069	0,3	0,023	0,2	0,008	0,1	0,003	0,1	0,001	0,1						
0,14	2,588	1,4	0,843	0,9	0,290	0,6	0,090	0,3	0,031	0,2	0,010	0,1	0,003	0,1	0,002	0,1						
0,16	3,285	1,6	1,068	1,0	0,367	0,6	0,114	0,4	0,039	0,2	0,013	0,2	0,004	0,1	0,002	0,1						
0,18	4,056	1,8	1,316	1,1	0,452	0,7	0,140	0,4	0,048	0,3	0,016	0,2	0,005	0,1	0,002	0,1	0,001	0,1				
0,20	4,900	2,0	1,588	1,2	0,544	0,8	0,168	0,5	0,058	0,3	0,019	0,2	0,006	0,1	0,003	0,1	0,001	0,1				
0,30	10,182	2,9	3,277	1,8	1,118	1,2	0,345	0,7	0,118	0,5	0,040	0,3	0,013	0,2	0,006	0,1	0,002	0,1	0,001	0,1		
0,40			5,499	2,5	1,868	1,6	0,574	1,0	0,196	0,6	0,066	0,4	0,022	0,2	0,010	0,2	0,004	0,1	0,002	0,1		
0,50			8,236	3,1	2,786	2,0	0,854	1,2	0,290	0,8	0,097	0,5	0,032	0,3	0,014	0,2	0,006	0,2	0,002	0,1		
0,60					3,869	2,4	1,183	1,4	0,401	0,9	0,134	0,6	0,045	0,4	0,020	0,3	0,008	0,2	0,003	0,1		
0,70					5,112	2,8	1,558	1,7	0,528	1,1	0,176	0,7	0,058	0,4	0,026	0,3	0,011	0,2	0,004	0,1		
0,80					6,513	3,1	1,980	1,9	0,669	1,2	0,223	0,8	0,074	0,5	0,032	0,3	0,014	0,2	0,005	0,2	0,003	0,1
0,90					8,071	3,5	2,448	2,2	0,826	1,4	0,275	0,9	0,091	0,6	0,040	0,4	0,017	0,3	0,006	0,2	0,003	0,1
1,00							2,960	2,4	0,997	1,5	0,332	1,0	0,110	0,6	0,048	0,4	0,020	0,3	0,008	0,2	0,004	0,2
1,20							4,117	2,9	1,382	1,8	0,459	1,2	0,152	0,7	0,066	0,5	0,028	0,4	0,011	0,2	0,006	0,2
1,40							5,449	3,4	1,824	2,1	0,604	1,4	0,199	0,9	0,087	0,6	0,037	0,4	0,014	0,3	0,007	0,2
1,60									2,322	2,5	0,767	1,6	0,253	1,0	0,110	0,7	0,046	0,5	0,018	0,3	0,009	0,3
1,80									2,874	2,8	0,948	1,7	0,311	1,1	0,136	0,8	0,057	0,5	0,022	0,4	0,011	0,3
2,00									3,480	3,1	1,145	1,9	0,376	1,2	0,164	0,9	0,069	0,6	0,026	0,4	0,014	0,3
2,20									4,139	3,4	1,360	2,1	0,446	1,3	0,194	1,0	0,081	0,7	0,031	0,4	0,016	0,3
2,40											1,591	2,3	0,521	1,5	0,227	1,0	0,095	0,7	0,036	0,5	0,019	0,4
2,60											1,839	2,5	0,601	1,6	0,261	1,1	0,109	0,8	0,041	0,5	0,021	0,4
2,80											2,104	2,7	0,686	1,7	0,298	1,2	0,125	0,8	0,047	0,6	0,024	0,4
3,00											2,385	2,9	0,777	1,8	0,337	1,3	0,141	0,9	0,053	0,6	0,027	0,5
3,20											2,682	3,1	0,873	2,0	0,379	1,4	0,158	1,0	0,060	0,6	0,031	0,5
3,40											2,995	3,3	0,974	2,1	0,422	1,5	0,176	1,0	0,067	0,7	0,035	0,5
3,60											3,324	3,5	1,080	2,2	0,468	1,6	0,195	1,1	0,074	0,7	0,039	0,6
3,80													1,190	2,3	0,515	1,6	0,215	1,1	0,081	0,8	0,043	0,6
4,00													1,306	2,4	0,565	1,7	0,235	1,2	0,089	0,8	0,047	0,6
4,20													1,427	2,6	0,617	1,8	0,257	1,3	0,097	0,8	0,051	0,7
4,40													1,553	2,7	0,671	1,9	0,279	1,3	0,105	0,9	0,055	0,7
4,60													1,683	2,8	0,727	2,0	0,302	1,4	0,114	0,9	0,059	0,7
4,80													1,819	2,9	0,785	2,1	0,326	1,4	0,123	1,0	0,064	0,7
5,00													1,959	3,1	0,845	2,2	0,351	1,5	0,132	1,0	0,069	0,8
5,20															0,895	2,3	0,373	1,6	0,138	1,0	0,073	0,8
5,40															0,962	2,3	0,399	1,6	0,151	1,1	0,079	0,8
5,60															1,030	2,4	0,426	1,7	0,161	1,1	0,084	0,9
5,80															1,093	2,5	0,454	1,8	0,171	1,2	0,091	0,9
6,00															1,166	2,6	0,483	1,8	0,182	1,2	0,096	0,9
6,20															1,241	2,7	0,512	1,9	0,193	1,3	0,102	1,0
6,40															1,310	2,8	0,542	1,9	0,204	1,3	0,108	1,0
6,60															1,389	2,9	0,574	2,0	0,216	1,3	0,114	1,0
6,80															1,470	3,0	0,605	2,1	0,227	1,4	0,120	1,1
7,00															1,544	3,0	0,638	2,1	0,240	1,4	0,126	1,1

Pressure Loss Tables

PPR S 3,2 (PN16) water temperature = 50° C

k=0,01	16 × 2,3 mm		20 × 2,8 mm		25 × 3,5 mm		32 × 4,5 mm		40 × 5,6 mm		50 × 6,9 mm		63 × 8,7 mm		75 × 10,4 mm		90 × 12,5 mm		110 × 15,2 mm		125 × 17,1 mm	
Q l/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s
0,02	0,068	0,2	0,022	0,1	0,008	0,1																
0,04	0,230	0,4	0,075	0,2	0,026	0,2	0,008	0,1	0,003	0,1												
0,06	0,473	0,6	0,154	0,4	0,053	0,2	0,016	0,1	0,006	0,1	0,002	0,1										
0,08	0,792	0,8	0,257	0,5	0,088	0,3	0,027	0,2	0,009	0,1	0,003	0,1										
0,10	1,183	1,0	0,382	0,6	0,131	0,4	0,040	0,2	0,014	0,2	0,005	0,1	0,002	0,1								
0,12	1,644	1,2	0,530	0,7	0,181	0,5	0,056	0,3	0,019	0,2	0,006	0,1	0,002	0,1	0,001	0,1						
0,14	2,175	1,4	0,698	0,9	0,238	0,6	0,073	0,3	0,025	0,2	0,008	0,1	0,003	0,1	0,001	0,1						
0,16	2,773	1,6	0,888	1,0	0,302	0,6	0,093	0,4	0,032	0,2	0,011	0,2	0,004	0,1	0,002	0,1						
0,18	3,439	1,8	1,099	1,1	0,373	0,7	0,115	0,4	0,039	0,3	0,013	0,2	0,004	0,1	0,002	0,1	0,001	0,1				
0,20	4,172	2,0	1,330	1,2	0,450	0,8	0,138	0,5	0,047	0,3	0,016	0,2	0,005	0,1	0,002	0,1	0,001	0,1				
0,30	8,828	2,9	2,785	1,8	0,935	1,2	0,285	0,7	0,096	0,5	0,032	0,3	0,011	0,2	0,005	0,1	0,002	0,1	0,001	0,1		
0,40			4,731	2,5	1,578	1,6	0,478	1,0	0,161	0,6	0,054	0,4	0,018	0,2	0,008	0,2	0,003	0,1	0,001	0,1		
0,50			7,161	3,1	2,376	2,0	0,716	1,2	0,240	0,8	0,080	0,5	0,026	0,3	0,012	0,2	0,005	0,2	0,002	0,1		
0,60					3,325	2,4	0,997	1,4	0,334	0,9	0,110	0,6	0,036	0,4	0,016	0,3	0,007	0,2	0,003	0,1		
0,70					4,425	2,8	1,322	1,7	0,441	1,1	0,146	0,7	0,048	0,4	0,021	0,3	0,009	0,2	0,003	0,1		
0,80					5,675	3,1	1,689	1,9	0,562	1,2	0,185	0,8	0,061	0,5	0,026	0,3	0,011	0,2	0,004	0,2	0,002	0,1
0,90					7,073	3,5	2,098	2,2	0,696	1,4	0,229	0,9	0,075	0,6	0,033	0,4	0,014	0,3	0,005	0,2	0,003	0,1
1,00							2,549	2,4	0,843	1,5	0,277	1,0	0,091	0,6	0,039	0,4	0,016	0,3	0,006	0,2	0,003	0,1
1,20							3,577	2,9	1,178	1,8	0,385	1,2	0,126	0,7	0,055	0,5	0,023	0,4	0,009	0,2	0,005	0,2
1,40							4,770	3,4	1,565	2,1	0,510	1,4	0,166	0,9	0,072	0,6	0,030	0,4	0,011	0,3	0,006	0,2
1,60									2,004	2,5	0,650	1,6	0,211	1,0	0,091	0,7	0,038	0,5	0,014	0,3	0,008	0,3
1,80									2,494	2,8	0,807	1,7	0,261	1,1	0,113	0,8	0,047	0,5	0,018	0,4	0,009	0,3
2,00									3,036	3,1	0,980	1,9	0,316	1,2	0,136	0,9	0,057	0,6	0,021	0,4	0,011	0,3
2,20									3,629	3,4	1,168	2,1	0,376	1,3	0,162	1,0	0,067	0,7	0,025	0,4	0,013	0,3
2,40											1,372	2,3	0,441	1,5	0,190	1,0	0,079	0,7	0,030	0,5	0,015	0,4
2,60											1,592	2,5	0,511	1,6	0,220	1,1	0,091	0,8	0,034	0,5	0,018	0,4
2,80											1,828	2,7	0,585	1,7	0,251	1,2	0,104	0,8	0,039	0,6	0,020	0,5
3,00											2,079	2,9	0,664	1,8	0,285	1,3	0,118	0,9	0,044	0,6	0,023	0,5
3,20											2,345	3,1	0,748	2,0	0,320	1,4	0,132	1,0	0,050	0,6	0,025	0,5
3,40											2,627	3,3	0,837	2,1	0,358	1,5	0,148	1,0	0,055	0,7	0,029	0,6
3,60											2,925	3,5	0,930	2,2	0,398	1,6	0,164	1,1	0,061	0,7	0,032	0,6
3,80													1,028	2,3	0,439	1,6	0,181	1,1	0,067	0,8	0,035	0,6
4,00													1,131	2,4	0,483	1,7	0,198	1,2	0,074	0,8	0,039	0,7
4,20													1,239	2,6	0,528	1,8	0,217	1,3	0,081	0,8	0,042	0,7
4,40													1,351	2,7	0,575	1,9	0,236	1,3	0,088	0,9	0,046	0,7
4,60													1,468	2,8	0,624	2,0	0,256	1,4	0,095	0,9	0,050	0,7
4,80													1,589	2,9	0,676	2,1	0,277	1,4	0,103	1,0	0,053	0,8
5,00													1,716	3,1	0,729	2,2	0,298	1,5	0,111	1,0	0,057	0,8
5,20															0,774	2,3	0,318	1,6	0,117	1,0	0,062	0,8
5,40															0,832	2,3	0,341	1,6	0,127	1,1	0,066	0,9
5,60															0,893	2,4	0,365	1,7	0,136	1,1	0,070	0,9
5,80															0,949	2,5	0,389	1,8	0,145	1,2	0,076	0,9
6,00															1,014	2,6	0,414	1,8	0,154	1,2	0,081	0,9
6,20															1,081	2,7	0,440	1,9	0,164	1,3	0,086	1,0
6,40															1,142	2,8	0,467	1,9	0,173	1,3	0,091	1,0
6,60															1,212	2,9	0,494	2,0	0,183	1,3	0,096	1,0
6,80															1,285	3,0	0,522	2,1	0,194	1,4	0,101	1,1
7,00															1,351	3,0	0,551	2,1	0,204	1,4	0,107	1,1

Pressure Loss Tables

PPR S 2,5 (PN20) water temperature = 10° C

k=0,01	16 × 2,7 mm		20 × 3,4 mm		25 × 4,2 mm		32 × 5,4 mm		40 × 6,7 mm		50 × 8,4 mm		63 × 10,5 mm		75 × 12,5 mm		90 × 15,0 mm		110 × 18,4 mm		125 × 20,8 mm	
Q l/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s
0,02	0,118	0,2	0,041	0,1	0,014	0,1	0,004	0,1														
0,04	0,399	0,5	0,140	0,3	0,047	0,2	0,015	0,1	0,005	0,1												
0,06	0,816	0,7	0,286	0,4	0,096	0,3	0,030	0,2	0,010	0,1	0,004	0,1										
0,08	1,357	0,9	0,475	0,6	0,159	0,4	0,050	0,2	0,017	0,1	0,006	0,1	0,002	0,1								
0,10	2,017	1,1	0,704	0,7	0,236	0,5	0,073	0,3	0,025	0,2	0,009	0,1	0,003	0,1	0,001	0,1						
0,12	2,791	1,4	0,973	0,9	0,325	0,6	0,101	0,3	0,034	0,2	0,012	0,1	0,004	0,1	0,002	0,1						
0,14	3,676	1,6	1,279	1,0	0,427	0,6	0,133	0,4	0,045	0,3	0,016	0,2	0,005	0,1	0,002	0,1	0,001	0,0				
0,16	4,669	1,8	1,622	1,2	0,540	0,7	0,168	0,5	0,057	0,3	0,020	0,2	0,006	0,1	0,003	0,1	0,001	0,1				
0,18	5,768	2,0	2,000	1,3	0,665	0,8	0,206	0,5	0,070	0,3	0,024	0,2	0,008	0,1	0,003	0,1	0,001	0,1				
0,20	6,971	2,3	2,414	1,5	0,802	0,9	0,249	0,6	0,084	0,4	0,029	0,2	0,010	0,1	0,004	0,1	0,002	0,1				
0,30	14,522	3,4	4,994	2,2	1,650	1,4	0,510	0,8	0,172	0,5	0,060	0,3	0,019	0,2	0,008	0,2	0,004	0,1	0,001	0,1		
0,40			8,397	2,9	2,761	1,8	0,849	1,1	0,286	0,7	0,099	0,5	0,032	0,3	0,014	0,2	0,006	0,1	0,002	0,1		
0,50					4,125	2,3	1,264	1,4	0,425	0,9	0,147	0,6	0,048	0,4	0,021	0,3	0,009	0,2	0,003	0,1		
0,60					5,735	2,8	1,752	1,7	0,587	1,1	0,203	0,7	0,066	0,4	0,029	0,3	0,012	0,2	0,005	0,1		
0,70					7,585	3,2	2,311	2,0	0,773	1,3	0,267	0,8	0,087	0,5	0,038	0,4	0,016	0,2	0,006	0,2		
0,80							2,939	2,3	0,981	1,4	0,338	0,9	0,110	0,6	0,048	0,4	0,020	0,3	0,008	0,2	0,004	0,2
0,90							3,635	2,5	1,211	1,6	0,417	1,0	0,135	0,6	0,059	0,5	0,025	0,3	0,010	0,2	0,005	0,2
1,00							4,399	2,8	1,463	1,8	0,503	1,2	0,163	0,7	0,071	0,5	0,030	0,4	0,011	0,2	0,006	0,2
1,20							6,127	3,4	2,031	2,2	0,696	1,4	0,225	0,9	0,097	0,6	0,041	0,4	0,016	0,3	0,008	0,2
1,40									2,683	2,5	0,917	1,6	0,296	1,0	0,128	0,7	0,054	0,5	0,021	0,3	0,011	0,3
1,60									3,417	2,9	1,165	1,8	0,375	1,2	0,162	0,8	0,068	0,6	0,026	0,4	0,013	0,3
1,80									4,233	3,2	1,441	2,1	0,463	1,3	0,200	0,9	0,083	0,6	0,032	0,4	0,017	0,3
2,00											1,742	2,3	0,559	1,4	0,241	1,0	0,101	0,7	0,039	0,5	0,021	0,4
2,20											2,070	2,5	0,663	1,6	0,286	1,1	0,119	0,8	0,046	0,5	0,024	0,4
2,40											2,423	2,8	0,775	1,7	0,334	1,2	0,139	0,8	0,054	0,6	0,028	0,4
2,60											2,803	3,0	0,894	1,9	0,385	1,3	0,160	0,9	0,062	0,6	0,033	0,5
2,80											3,208	3,2	1,022	2,0	0,440	1,4	0,183	1,0	0,070	0,7	0,037	0,5
3,00											3,638	3,5	1,158	2,2	0,498	1,5	0,207	1,1	0,080	0,7	0,042	0,6
3,20													1,301	2,3	0,559	1,6	0,232	1,1	0,089	0,8	0,047	0,6
3,40													1,452	2,5	0,623	1,7	0,259	1,2	0,099	0,8	0,052	0,6
3,60													1,610	2,6	0,691	1,8	0,286	1,3	0,110	0,9	0,058	0,7
3,80													1,776	2,7	0,761	1,9	0,316	1,3	0,121	0,9	0,064	0,7
4,00													1,949	2,9	0,835	2,0	0,346	1,4	0,133	1,0	0,069	0,7
4,20													2,131	3,0	0,912	2,1	0,377	1,5	0,145	1,0	0,076	0,8
4,40													2,319	3,2	0,992	2,2	0,410	1,6	0,157	1,0	0,083	0,8
4,60													2,515	3,3	1,075	2,3	0,444	1,6	0,170	1,1	0,089	0,8
4,80													2,718	3,5	1,161	2,4	0,480	1,7	0,184	1,1	0,097	0,9
5,00															1,251	2,5	0,516	1,8	0,198	1,2	0,105	0,9
5,20															1,332	2,7	0,548	1,8	0,207	1,2	0,111	1,0
5,40															1,426	2,8	0,587	1,9	0,222	1,3	0,120	1,0
5,60															1,522	2,9	0,626	2,0	0,235	1,3	0,128	1,0
5,80															1,622	3,0	0,667	2,1	0,251	1,4	0,135	1,1
6,00															1,735	3,1	0,710	2,1	0,268	1,4	0,145	1,1
6,20																	0,753	2,2	0,285	1,5	0,152	1,1
6,40																	0,797	2,3	0,300	1,5	0,162	1,2
6,60																	0,843	2,3	0,318	1,6	0,172	1,2
6,80																	0,897	2,4	0,336	1,6	0,179	1,2
7,00																	0,945	2,5	0,352	1,7	0,190	1,3

Pressure Loss Tables

PPR S 2,5 (PN20) water temperature = 50° C

k=0,01	16 × 2,7mm		20 × 3,4mm		25 × 4,2mm		32 × 5,4mm		40 × 6,7mm		50 × 8,4mm		63 × 10,5mm		75 × 12,5mm		90 × 15,0mm		110 × 18,4mm		125 × 20,8mm	
Q l/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s
0,02	0,096	0,2	0,034	0,1	0,011	0,1	0,004	0,1														
0,04	0,326	0,5	0,114	0,3	0,038	0,2	0,012	0,1	0,004	0,1												
0,06	0,672	0,7	0,234	0,4	0,078	0,3	0,024	0,2	0,008	0,1	0,003	0,1										
0,08	1,126	0,9	0,390	0,6	0,130	0,4	0,040	0,2	0,014	0,1	0,005	0,1	0,002	0,1								
0,10	1,684	1,1	0,582	0,7	0,193	0,5	0,060	0,3	0,020	0,2	0,007	0,1	0,002	0,1	0,001	0,1						
0,12	2,344	1,4	0,807	0,9	0,267	0,6	0,082	0,3	0,028	0,2	0,010	0,1	0,003	0,1	0,001	0,1						
0,14	3,104	1,6	1,065	1,0	0,351	0,6	0,108	0,4	0,037	0,3	0,013	0,2	0,004	0,1	0,002	0,1	0,001	0,0				
0,16	3,962	1,8	1,356	1,2	0,446	0,7	0,137	0,5	0,046	0,3	0,016	0,2	0,005	0,1	0,002	0,1	0,001	0,1				
0,18	4,918	2,0	1,679	1,3	0,551	0,8	0,169	0,5	0,057	0,3	0,020	0,2	0,006	0,1	0,003	0,1	0,001	0,1				
0,20	5,972	2,3	2,033	1,5	0,666	0,9	0,204	0,6	0,069	0,4	0,024	0,2	0,008	0,1	0,003	0,1	0,001	0,1				
0,30	12,680	3,4	4,273	2,2	1,388	1,4	0,423	0,8	0,141	0,5	0,049	0,3	0,016	0,2	0,007	0,2	0,003	0,1	0,001	0,1		
0,40			7,281	2,9	2,348	1,8	0,710	1,1	0,236	0,7	0,081	0,5	0,026	0,3	0,011	0,2	0,005	0,1	0,002	0,1		
0,50					3,541	2,3	1,065	1,4	0,353	0,9	0,121	0,6	0,039	0,4	0,017	0,3	0,007	0,2	0,003	0,1		
0,60					4,964	2,8	1,486	1,7	0,491	1,1	0,168	0,7	0,054	0,4	0,023	0,3	0,010	0,2	0,004	0,1		
0,70					6,616	3,2	1,972	2,0	0,649	1,3	0,221	0,8	0,071	0,5	0,031	0,4	0,013	0,2	0,005	0,2		
0,80							2,523	2,3	0,828	1,4	0,281	0,9	0,090	0,6	0,039	0,4	0,016	0,3	0,006	0,2	0,003	0,2
0,90							3,138	2,5	1,027	1,6	0,348	1,0	0,111	0,6	0,048	0,5	0,020	0,3	0,008	0,2	0,004	0,2
1,00							3,816	2,8	1,245	1,8	0,421	1,2	0,135	0,7	0,058	0,5	0,024	0,4	0,009	0,2	0,005	0,2
1,20							5,364	3,4	1,742	2,2	0,587	1,4	0,187	0,9	0,080	0,6	0,033	0,4	0,013	0,3	0,007	0,2
1,40									2,317	2,5	0,778	1,6	0,247	1,0	0,106	0,7	0,044	0,5	0,017	0,3	0,009	0,3
1,60									2,971	2,9	0,994	1,8	0,315	1,2	0,135	0,8	0,056	0,6	0,021	0,4	0,011	0,3
1,80									3,702	3,2	1,235	2,1	0,390	1,3	0,167	0,9	0,069	0,6	0,026	0,4	0,014	0,3
2,00											1,501	2,3	0,473	1,4	0,202	1,0	0,083	0,7	0,032	0,5	0,017	0,4
2,20											1,791	2,5	0,563	1,6	0,240	1,1	0,099	0,8	0,038	0,5	0,019	0,4
2,40											2,106	2,8	0,660	1,7	0,281	1,2	0,116	0,8	0,044	0,6	0,023	0,4
2,60											2,445	3,0	0,765	1,9	0,325	1,3	0,134	0,9	0,051	0,6	0,027	0,5
2,80											2,809	3,2	0,877	2,0	0,373	1,4	0,153	1,0	0,058	0,7	0,030	0,5
3,00											3,197	3,5	0,996	2,2	0,423	1,5	0,174	1,1	0,066	0,7	0,035	0,6
3,20													1,123	2,3	0,476	1,6	0,195	1,1	0,074	0,8	0,039	0,6
3,40													1,256	2,5	0,532	1,7	0,218	1,2	0,083	0,8	0,043	0,6
3,60													1,397	2,6	0,591	1,8	0,242	1,3	0,092	0,9	0,048	0,7
3,80													1,545	2,7	0,653	1,9	0,267	1,3	0,101	0,9	0,054	0,7
4,00													1,701	2,9	0,718	2,0	0,293	1,4	0,111	1,0	0,058	0,7
4,20													1,863	3,0	0,786	2,1	0,321	1,5	0,121	1,0	0,064	0,8
4,40													2,033	3,2	0,856	2,2	0,349	1,6	0,132	1,0	0,070	0,8
4,60													2,210	3,3	0,930	2,3	0,379	1,6	0,143	1,1	0,075	0,8
4,80													2,394	3,5	1,006	2,4	0,410	1,7	0,155	1,1	0,081	0,9
5,00															1,086	2,5	0,442	1,8	0,167	1,2	0,088	0,9
5,20															1,158	2,7	0,470	1,8	0,175	1,2	0,093	1,0
5,40															1,242	2,8	0,504	1,9	0,188	1,3	0,101	1,0
5,60															1,327	2,8	0,539	2,0	0,199	1,3	0,108	1,0
5,80															1,416	2,9	0,575	2,1	0,214	1,4	0,114	1,1
6,00															1,517	3,1	0,612	2,1	0,228	1,4	0,122	1,1
6,20																	0,651	2,2	0,243	1,5	0,128	1,1
6,40																	0,690	2,3	0,256	1,5	0,137	1,2
6,60																	0,730	2,3	0,272	1,6	0,146	1,2
6,80																	0,778	2,4	0,288	1,6	0,152	1,2
7,00																	0,821	2,5	0,301	1,7	0,162	1,3

Pressure Loss Tables

PPR S 2,5 (PN20) water temperature = 80° C

k=0,01	16 × 2,7mm		20 × 3,4mm		25 × 4,2mm		32 × 5,4mm		40 × 6,7mm		50 × 8,4mm		63 × 10,5mm		75 × 12,5mm		90 × 15,0mm		110 × 18,4mm		125 × 20,8mm	
Q l/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s	R kPa/m	v m/s
0,02	0,087	0,2	0,030	1,1	0,010	0,1	0,003	0,1														
0,04	0,299	0,5	0,104	0,3	0,035	0,2	0,011	0,1	0,004	0,1												
0,06	0,619	0,7	0,214	0,4	0,071	0,3	0,022	0,2	0,007	0,1	0,003	0,1										
0,08	1,042	0,9	0,359	0,6	0,119	0,4	0,037	0,2	0,012	0,1	0,004	0,1	0,001	0,1								
0,10	1,565	1,1	0,536	0,7	0,177	0,5	0,054	0,3	0,018	0,2	0,006	0,1	0,002	0,1	0,001	0,1						
0,12	2,186	1,4	0,746	0,9	0,245	0,6	0,075	0,3	0,025	0,2	0,009	0,1	0,003	0,1	0,001	0,1						
0,14	2,905	1,6	0,988	1,0	0,323	0,6	0,099	0,4	0,033	0,3	0,012	0,2	0,004	0,1	0,002	0,1	0,001	0,0				
0,16	3,719	1,8	1,261	1,2	0,412	0,7	0,126	0,5	0,042	0,3	0,015	0,2	0,005	0,1	0,002	0,1	0,001	0,1				
0,18	4,630	2,0	1,565	1,3	0,510	0,8	0,155	0,5	0,052	0,3	0,018	0,2	0,006	0,1	0,003	0,1	0,001	0,1				
0,20	5,636	2,3	1,900	1,5	0,617	0,9	0,188	0,6	0,063	0,4	0,022	0,2	0,007	0,1	0,003	0,1	0,001	0,1				
0,30	12,090	3,4	4,031	2,2	1,296	1,4	0,391	0,8	0,130	0,5	0,045	0,3	0,014	0,2	0,006	0,2	0,003	0,1	0,001	0,1		
0,40			6,918	2,9	2,206	1,8	0,661	1,1	0,218	0,7	0,075	0,5	0,024	0,3	0,010	0,2	0,004	0,1	0,002	0,1		
0,50					3,346	2,3	0,995	1,4	0,327	0,9	0,111	0,6	0,036	0,4	0,015	0,3	0,006	0,2	0,002	0,1		
0,60					4,712	2,8	1,395	1,7	0,456	1,1	0,155	0,7	0,050	0,4	0,021	0,3	0,009	0,2	0,003	0,1		
0,70					6,304	3,2	1,858	2,0	0,605	1,3	0,205	0,8	0,065	0,5	0,028	0,4	0,012	0,2	0,005	0,2		
0,80							2,384	2,3	0,774	1,4	0,261	0,9	0,083	0,6	0,036	0,4	0,015	0,3	0,006	0,2	0,003	0,2
0,90							2,974	2,5	0,963	1,6	0,324	1,0	0,103	0,6	0,044	0,5	0,018	0,3	0,007	0,2	0,003	0,2
1,00							3,626	2,8	1,171	1,8	0,392	1,2	0,124	0,7	0,053	0,5	0,022	0,4	0,009	0,2	0,004	0,2
1,20							5,121	3,4	1,645	2,2	0,549	1,4	0,173	0,9	0,074	0,6	0,031	0,4	0,012	0,3	0,006	0,2
1,40									2,197	2,5	0,730	1,6	0,230	1,0	0,098	0,7	0,040	0,5	0,016	0,3	0,008	0,3
1,60									2,826	2,9	0,936	1,8	0,293	1,2	0,125	0,8	0,051	0,6	0,020	0,4	0,010	0,3
1,80									3,532	3,2	1,166	2,1	0,364	1,3	0,155	0,9	0,064	0,6	0,024	0,4	0,012	0,3
2,00											1,421	2,3	0,443	1,4	0,188	1,0	0,077	0,7	0,029	0,5	0,015	0,4
2,20											1,700	2,5	0,528	1,6	0,224	1,1	0,092	0,8	0,035	0,5	0,018	0,4
2,40											2,003	2,8	0,621	1,7	0,263	1,2	0,107	0,8	0,041	0,6	0,021	0,4
2,60											2,331	3,0	0,721	1,9	0,304	1,3	0,124	0,9	0,047	0,6	0,024	0,5
2,80											2,682	3,2	0,828	2,0	0,349	1,4	0,142	1,0	0,054	0,7	0,027	0,5
3,00											3,058	3,5	0,942	2,2	0,397	1,5	0,162	1,1	0,061	0,7	0,031	0,6
3,20													1,064	2,3	0,447	1,6	0,182	1,1	0,069	0,8	0,036	0,6
3,40													1,192	2,5	0,501	1,7	0,204	1,2	0,077	0,8	0,039	0,6
3,60													1,328	2,6	0,557	1,8	0,226	1,3	0,085	0,9	0,044	0,7
3,80													1,471	2,7	0,616	1,9	0,250	1,3	0,094	0,9	0,049	0,7
4,00													1,621	2,9	0,679	2,0	0,275	1,4	0,103	1,0	0,053	0,7
4,20													1,778	3,0	0,744	2,1	0,301	1,5	0,113	1,0	0,058	0,8
4,40													1,942	3,2	0,812	2,2	0,328	1,6	0,123	1,0	0,064	0,8
4,60													2,113	3,3	0,882	2,3	0,356	1,6	0,134	1,1	0,068	0,8
4,80													2,292	3,5	0,956	2,4	0,386	1,7	0,145	1,1	0,074	0,9
5,00															1,033	2,5	0,416	1,8	0,156	1,2	0,081	0,9
5,20															1,081	2,7	0,436	1,8	0,161	1,2	0,085	1,0
5,40															1,160	2,8	0,467	1,9	0,173	1,3	0,092	1,0
5,60															1,242	2,9	0,500	2,0	0,184	1,3	0,099	1,0
5,80															1,326	3,0	0,534	2,1	0,199	1,4	0,105	1,1
6,00															1,422	3,1	0,569	2,1	0,210	1,4	0,112	1,1
6,20																	0,605	2,2	0,224	1,5	0,118	1,1
6,40																	0,642	2,3	0,236	1,5	0,126	1,2
6,60																	0,680	2,3	0,251	1,6	0,134	1,2
6,80																	0,725	2,4	0,266	1,6	0,140	1,2
7,00																	0,765	2,5	0,279	1,7	0,149	1,3

Fittings Range

Elbow 90 °



Elbow 45°



Tee



Reducing Tee



Socket



Union



Reducer Internal / External



End Cap



Elbow 90° with Metal Male Thread



Tee with Metal Female Thread



Tee with Metal Male Thread



Flange Adaptor



Reducing Sleeve with Metal Female Thread



Reducing Sleeve with Metal Male Thread



Elbow 90° with Metal Female Thread



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Jul.-2026